Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
Fall 2025

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Mutual Funds
(Chapter 20)

Outline of today's class

Mutual Funds (MFs)

1. What they are and why they have grown so much
2. What they do
3. Structure
4. Prices and Fees
5. Types of MFs

a. Money Markets Mutual Funds (MMMFs)
b. Index funds
c. Exchange Traded Funds (ETFs)
d. Hedge Funds
e. Conflicts of interests (optional)

Mutual Funds

- ✓ **Mutual Funds** (MFs) are financial intermediaries that:
  - pool the investors' resources by **selling** them **shares**

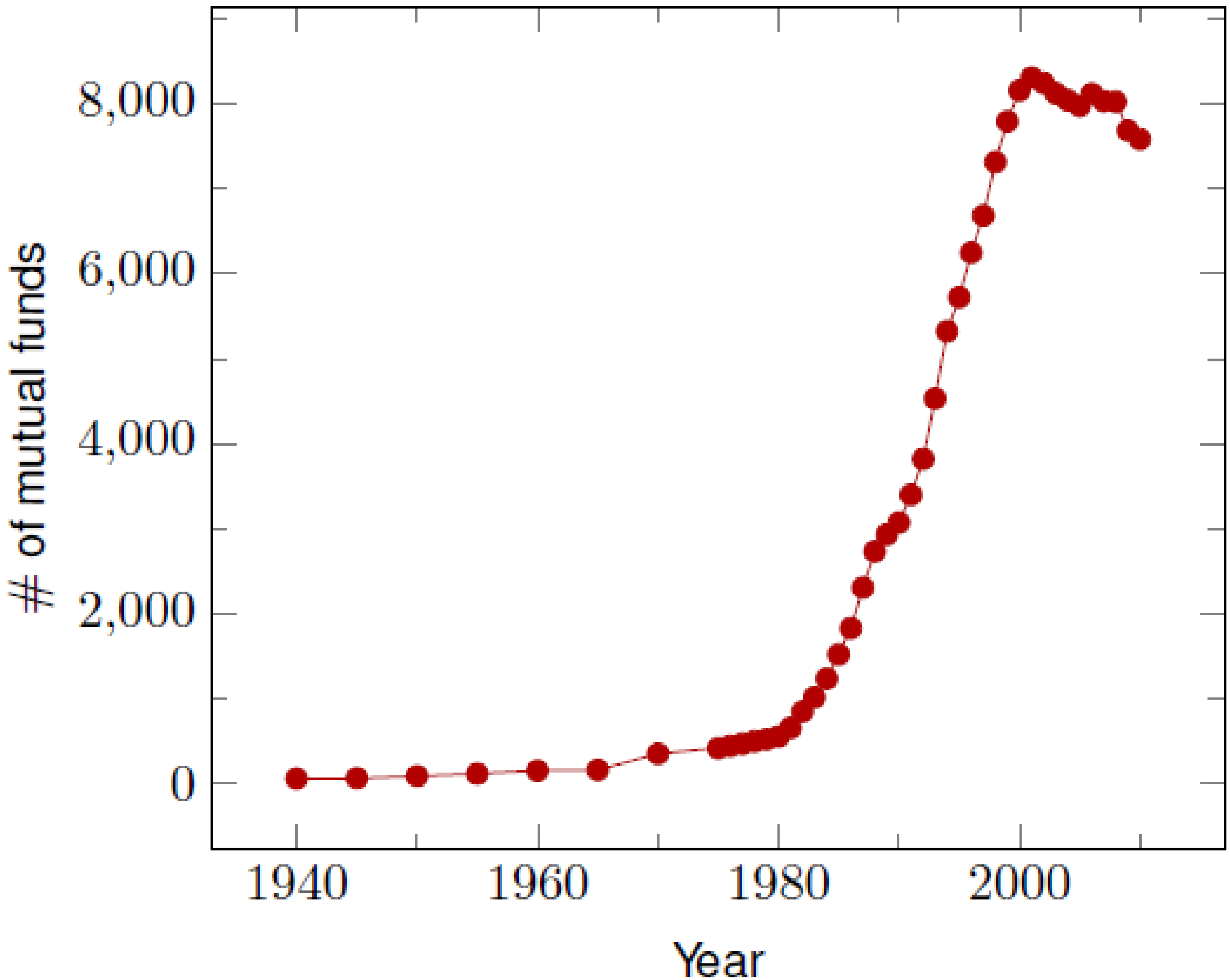
    (so, investors are shareholders)

    and
  - using the proceeds to invest in portfolios of securities

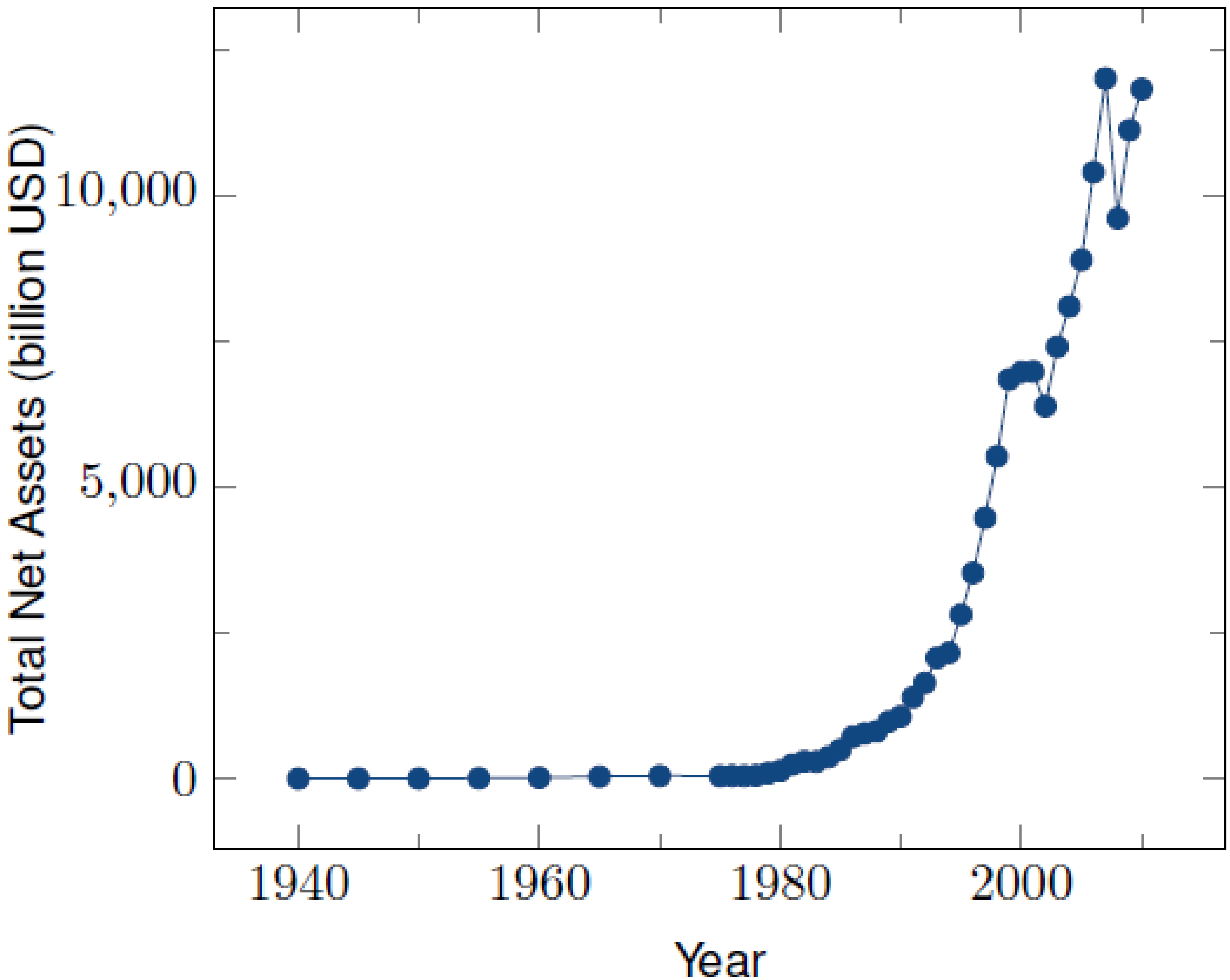
- ✓ They have been growing in number and importance since 1980, in the US and worldwide
  - due to ceiling on int. rates (regulation Q, until 1986):
    - inflation in the 70's made MFs attractive w.r.t. bank deposits
  - equity boom in the '80s and '90s made the rest

- ✓ MFs are now almost as large commercial banks in terms of *total assets* [**Asset Under Management** (AUM)]

The Growth of US Mutual Funds:
Number of Funds



The Growth of US Mutual Funds:
Assets Under Management (AUM)



Mutual Funds: What do they do?

1. BENEFITS: the *pooling* of resources
   - reduces the *transaction costs*

     and
   - brings better *diversification* to small investors

2. TYPES: the majority of MFs continuously
   - sells *new shares* to investors

     and
   - allows *redemption of outstanding shares* at market price

Benefits: Diversification, Denomination and Transaction Costs

- ✓ Denomination:
  - individual investors would like to diversify, but it's not always possible: with $1,000 or even $10,000, difficult to diversify across assets/sectors/countries ...
  - MFs pool individual savings and issue individual shares of low denomination (ex.: $1,000)

- ✓ Diversification benefits come with the ability to invest
  - in *large denomination* assets

    and
  - at *lower average* transaction costs

- ✓ But how do Mutual Funds make money?
  - they charge **fees** (more on this later)

The Benefits of Mutual Funds

5 main **benefits** of MFs explain why they're so popular

1. ***Liquidity* intermediation**: investors can *quickly* convert investments into *cash* at low or zero fee

All related
see Lecture 14

2. ***Denomination* intermediation**: investors can participate in equity/debt offerings that, individually, require more capital than they possess
3. ***Diversification***: even with small investments, investors can *diversify* to reduce risk
4. ***Cost* advantages**: MF can negotiate *lower transaction fees* than would be available to the individual investor
5. **Managerial expertise**: many investors prefer to rely on *professional money managers* to select their investments

Mutual Fund Structure

1. Closed-End Fund

- fixed number of nonredeemable shares
  - shares sold through an IPO …
  - … then traded in the OTC/Stock Exchange
- after IPO **no other $ can be added** nor **withdrawn**
  - so, investors **can't** redeem their shares …
  - … but they can sell them to *others* at market value: people know **Net Asset Value** (NAV) bcs composition is fixed, which is determined by supply and demand of each asset in the fund
- trade value
  - at substantial discounts relative to their NAV
  - but prices converge to reported NAV at termination
- termination
  - either through liquidation, merger or conversion to an open-end fund
- more common for funds that invest in *less liquid* securities
  - ex.: small firms, municipal bonds

Mutual Fund Structure

2. Open-End Fund

- investors *may* buy or redeem shares (ask for cash back) at any time
  - so fund can grow to +∞ …
- … but fund need to *keep some liquidity* …
  - … to be able to meet redemption requests at all times
- Both purchase and redemption (buy back by MF) take place at share price determined by the NAV of MF

✓ 98% of MFs assets are in open-end funds

Calculating a Mutual Fund's Net Asset Value

Net Asset Value (NAV) $(\sim \textit{sort of price})$

- *Definition:* Total market value of the mutual fund's stocks, bonds, cash, and other assets minus any liabilities (such as accrued fees), divided by the number of shares outstanding

Stocks	$35,000,000
Bonds	$15,000,000
Cash	$3,000,000
Total value of assets	$53,000,000
Liabilities	–$800,000
Net worth	$52,200,000
Outstanding shares	15 million

market values, calculated using market prices, not *historical purchase* (book) values

NAV = $52,200,000/15,000,000 = $3.48

$$NAV = \frac{Assets - Liabilities}{\#\ shares} = \frac{Net\ Worth}{\#\ shares}$$

Exercise: yield of MF

Suppose that after 1 day, values change as follows:

Stocks= \$ 34 M, Bonds = \$ 18 M, Cash = \$ 4 M, Liabilities= \$ 0.9 M.

Q1: What is new NAV?

Q2: What is the yield of the investment in the MF?

Exercise: yield of MF

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Q1: What is new NAV?

A1: NAV = (Assets - Liabilities)/# shares

NAV = (*\$55.2M – \$0.9M)/15M = \$3.7*

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Exercise: yield of MF

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NAV = (*\$55.2M – \$0.9M)/15M = \$3.7*

Q2: What is the yield of the investment in the MF?

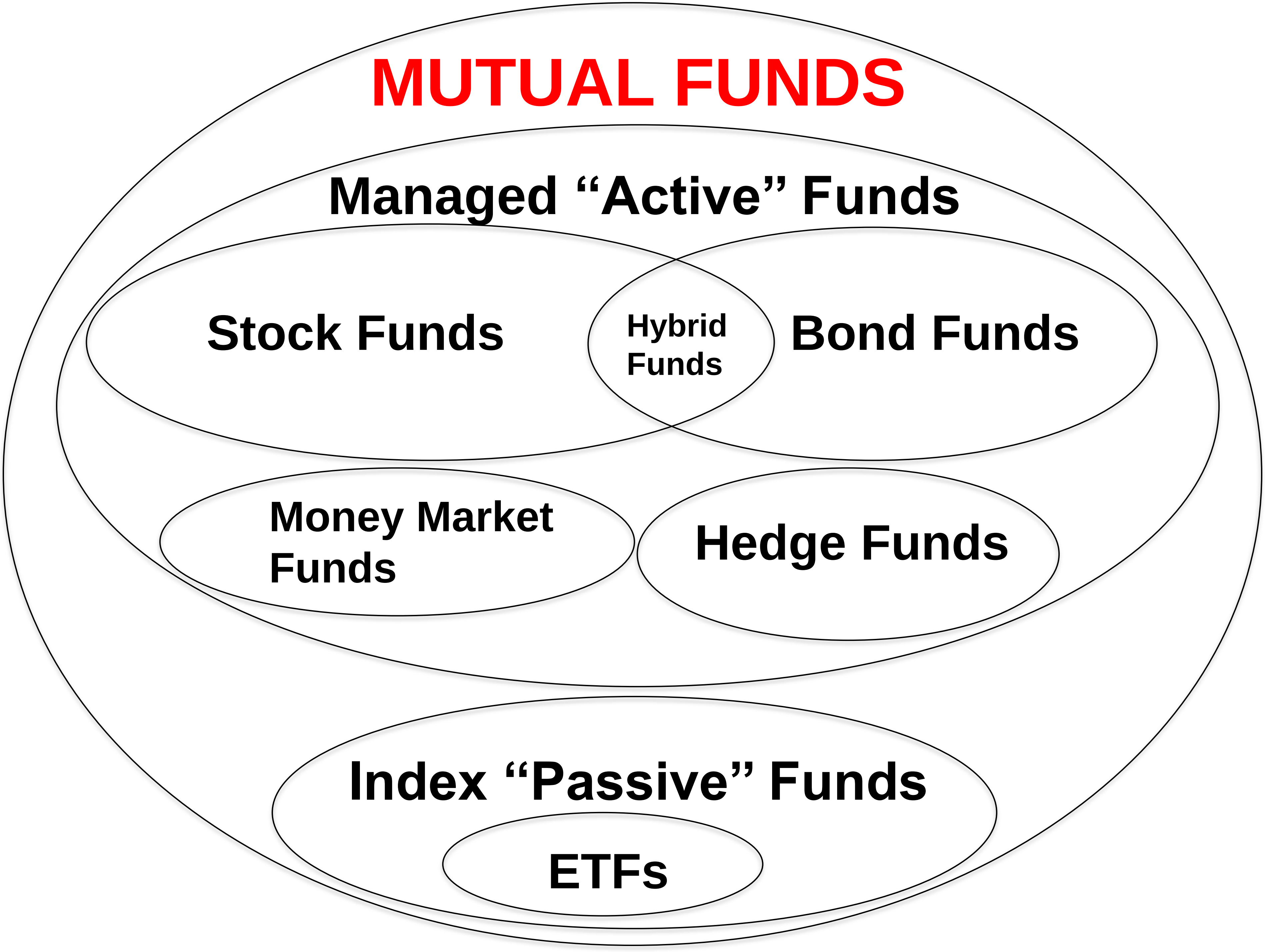
A2: The yield is the percentage change of NAV:

$$yield = \frac{\$3.7 - \$3.48}{\$3.48} = 5.6\%$$

Investment Objective Classes

- ✓ Four primary classes of MFs by *asset class* they invest in
  - ***Long term MFs***
    - *Stock* (equity) funds [active]
    - *Bond* funds [active]
    - *Hybrid* funds [active]
  - ***Short term MFs***
    - *Money market* funds [active]

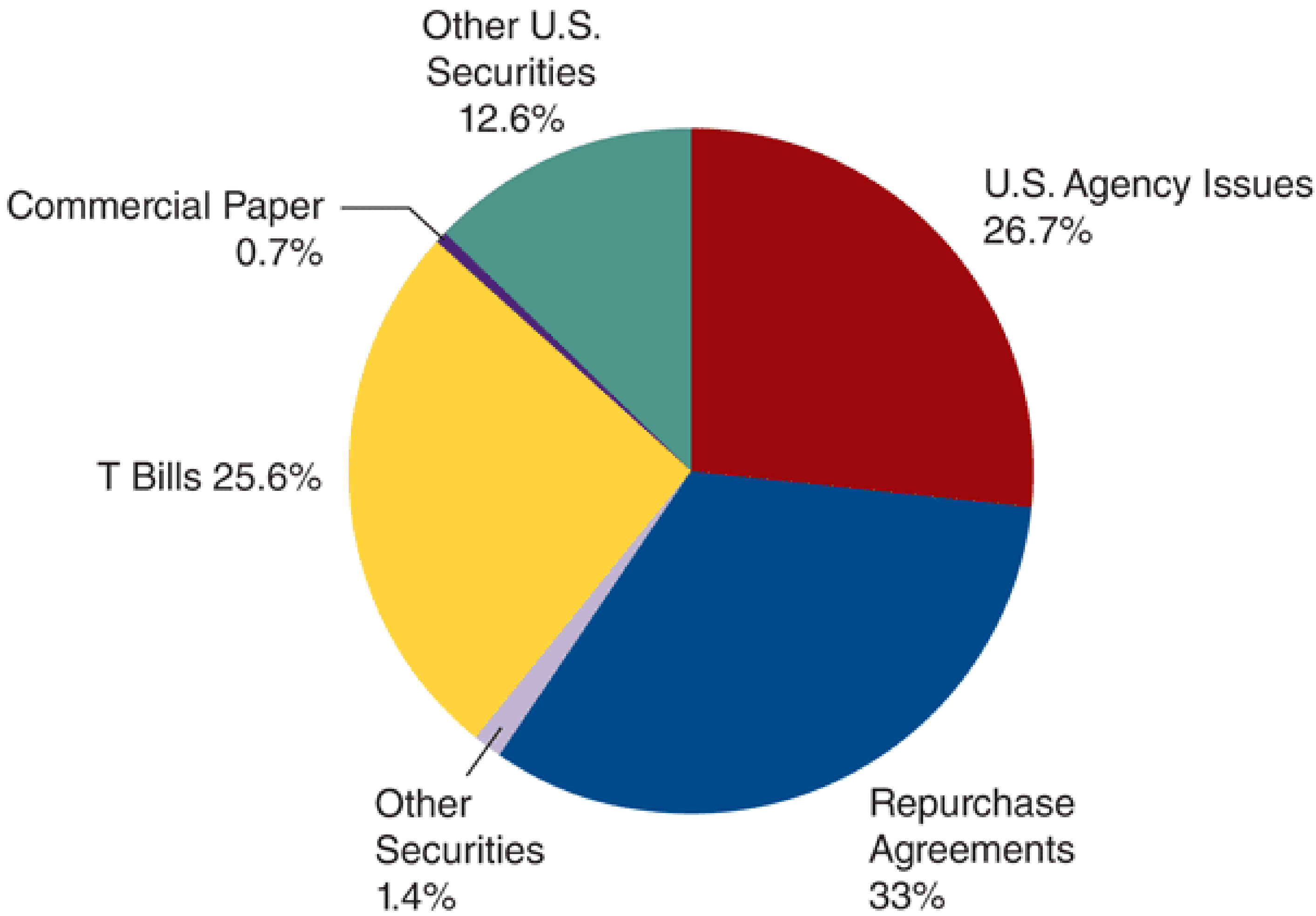
- ✓ *Other* types that do not fit in these categories:
  - ***Index* Funds** (subset: **ETFs**) [passive]
  - ***Hedge* Funds** [active]


MUTUAL FUNDS
Managed “Active” Funds
Stock Funds
Hybrid Funds
Bond Funds
Money Market Funds
Hedge Funds
Index “Passive” Funds
ETFs

Money Market Mutual Funds (MMMFs)

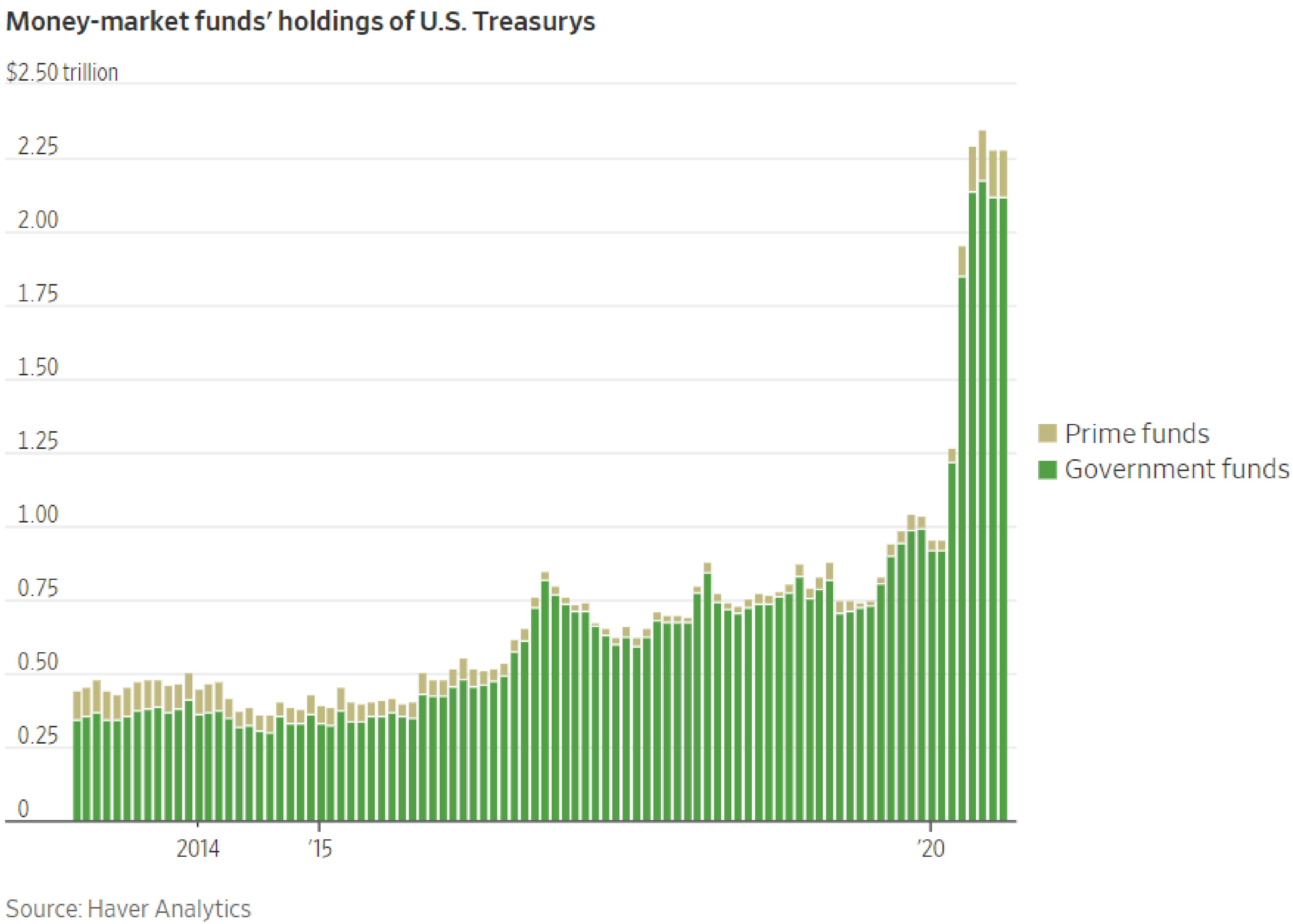
- ✓ Open-end funds that invest only in ***money*** *market* securities
- ✓ Traditionally thought to be <u>very safe</u>
  - they invest in money market instruments!
  - and yield *higher returns* than *bank deposits*
- ✓ That's why they are so popular and net assets increased dramatically since 1980
  - overcome interest rate ceilings on deposits due to Regulation Q
- ✓ *Low* denomination
  - initial minimum investment is usually in the range $500-$2,000
- ✓ Some of these MMMFs went into <u>trouble during the 2007 crisis</u> (saved by Treasury and Fed)
- ✓ What do they invest on? (next slide)

Average Distribution of MMMF Assets



Source: Investment Company Institute, *2012 Investment Company Fact Book*, 53rd ed. (Washington, DC: ICI), http://www.icifactbook.org/pdf/13_fb_table43.pdf.

MMMF after Covid



Fee Structure of Mutual Funds

MF often have **fees** (4 ex., there can be one-time fees on the amount of shares you purchase (≈2-6% value)

✓ **Load funds**

- charge an *upfront fee,* a one-time commission at the time of the purchase of the share (% of *total* sale)

✓ **Deferred load funds**

- charge a one-time fee when the shares are redeemed (% of *original* sale)

✓ **No Load funds**

- charge you no commission for purchasing / selling shares (but still charge other fees)
- this represents 55-65% of bond & equity funds

Is this all the fees? No! Other fees include:

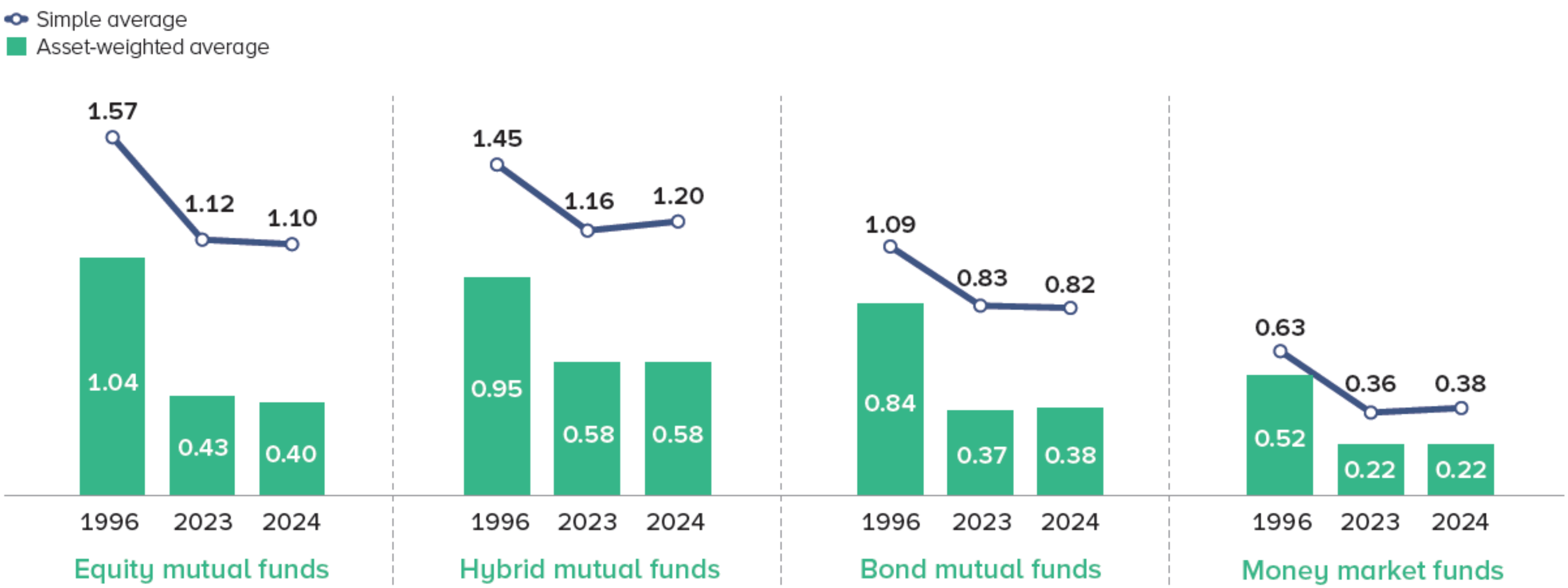
management fees, account maintenance fees, marketing fees, etc...

Fee Structure of Mutual Funds

FIGURE 1

Average Expense Ratios Incurred by Mutual Fund Investors Have Declined Substantially Since 1996

Percent



Note: For additional data, see Figure S1 in the statistical appendix.

Sources: Investment Company Institute, Lipper, and Morningstar

The *average expense ratio* for MFs is an annual fee, expressed as a percentage of investment, that covers the fund's operating costs (ex.: management, administrative, and marketing expenses)

Exercise: MF fees and Investing

Consider an MF with the following assets at the end of the trading day:

Asset	Units Owned	Market Price per security	Book Value per security
Stock A	10	$15	$20
Stock B	20	$20	$15
Stock C	15	$30	$25
Bond	10	$20	$49
Cash	--	$50	$50

The MF has 100 shares outstanding

Q1: What is the NAV of this MF?

Exercise: MF fees and Investing

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The MF has 100 shares outstanding

Q1: What is the NAV of this MF?

A1: $\mathbf{NAV} = (\$15 \times 10 + \$20 \times 20 + \$30 \times 15 + \$20 \times 10 + \$50)/100$

$\mathbf{NAV} = \$1{,}250/100 = \$\mathbf{12.5}$

Note that we used market prices, not book values

Exercise: MF fees and Investing

You decide to buy into this fund and **want to invest $25**

Q2: How many shares can you buy?

Q3: What is the total new number of shares in this MF and the new NAV?

Q4: If the fund charges a 5% upfront fee, how much money do you need to hand out to buy **$25 (net)** of shares?

Exercise: MF fees and Investing

You decide to buy into this fund and **want to invest $25**

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A2: You can buy 2 shares: if NAV = $12.5, $25/$12.5 = 2

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Q3: What is the total new number of shares in this MF and the new NAV?

A3: The new total number of MF shares is 102 (= 100 + 2). However the assets (cash) also increase by $25, so NAV is unchanged at $12.5:

$$NAV = \frac{\$1{,}250 + \$25}{100 + 2} = \frac{\$1{,}275}{102} = \$12.5$$

Q4: If the fund charges a 5% upfront fee, how much money do you need to hand out to buy **$25 (net)** of shares?

Exercise: MF fees and Investing

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Q4: If the fund charges a 5% upfront fee, how much money do you need to hand out to buy **$25 (net)** of shares?

A4: Now with $25 you buy only 95% of your *desired* number of shares; 5% is paid as fee. So, you need to give to the manager:

$$\$25{:}\,0.95 = x{:}\,1 \Rightarrow x = \frac{\$25}{0.95} = \$26.32$$

Index Funds

- ✓ A special class of mutual funds that is *not actively* managed
  - the fund **contains the stock of the index it is *mimicking***. 4 ex.: S&P500, Russell 1000, etc…
- ✓ Advantages
  - they offer benefits of traditional mutual funds (good diversification) …
  - with lower fees (passive tracking, no brainer)
- ✓ Only recently popular:
  - until the advent of computing technology, it was quite costly to manage an index portfolio of 100s or 1,000s stocks
- ✓ If you believe in *EMH*, buy passively managed funds: nobody can systematically beat the market
  - the former CEO of Vanguard Group admitted that he himself was an "index" investor

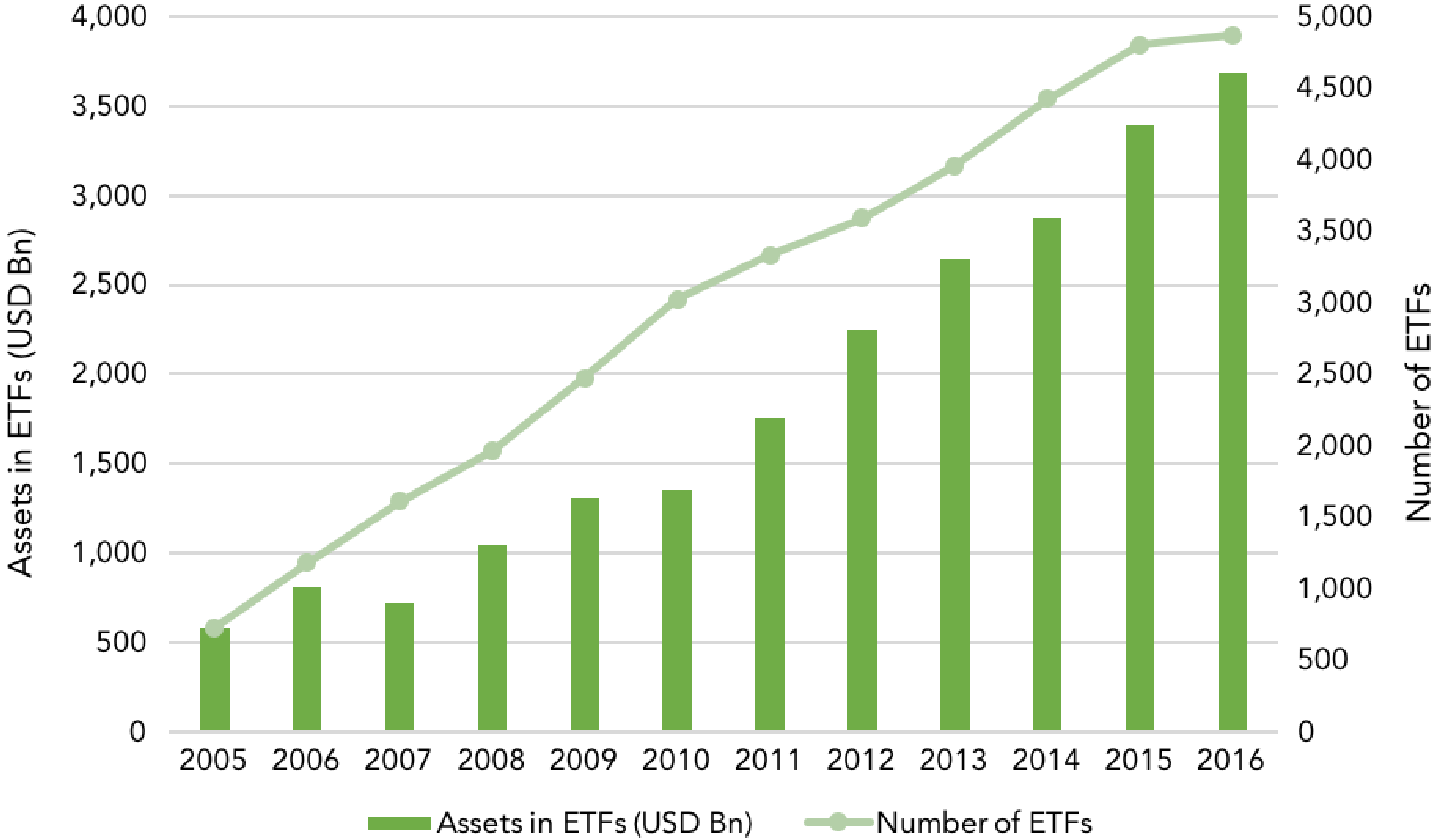
ETFs History

- ✓ **Exchange Traded Funds** (ETFs) are *open-end* investment companies that replicate an index

- ✓ One of the **most important innovations** in financial markets of the last decades

- ✓ History
  - January 1993: First US-listed ETF, the SPDR, launched by State Street
    - SPDR tracks S&P500 and today still largest ETF with AUM = $700 B (look)
  - after SPDR, new ETFs were launched tracking domestic and international indices, and more specialized sector, regions…
  - recently, ETFs have **grown substantially** in assets, diversity, and market significance

ETF Growth



4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
Assets in ETFs (USD Bn)
5,000
4,500
4,000
3,500
3,000
2,500
2,000
1,500
1,000
500
0
Number of ETFs
2005
2006
2007
2008
2009
2010
2011
2012
2013
2014
2015
2016
Assets in ETFs (USD Bn)
Number of ETFs

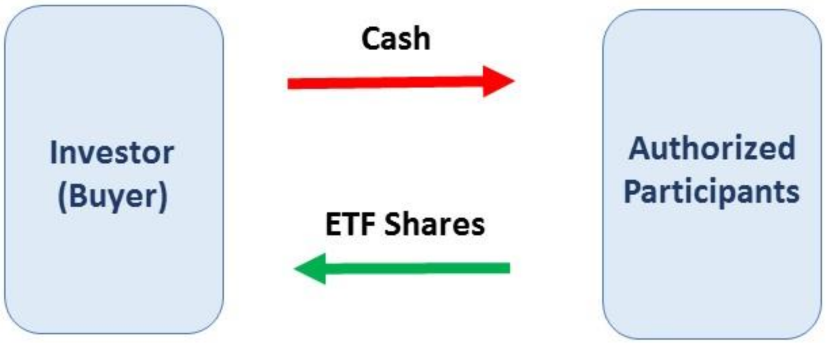
A Special Type of Index Funds: ETFs

- ✓ ETFs are a mix between Index funds and MFs

- ✓ Differently from (and better than) MFs, ETFs are:
  - more liquid than MFs
  - trade continuously during the trading day (like stocks)
    - open-end MFs transactions occur *at the end of the day* at NAV
    - closed-end MFs shares also trade *continuously* like ETFs, but they suffer from poor liquidity
  - lower fees than MFs
    - ETFs are *passively* managed (just track a benchmark, no-brainer), so have substantially lower fees than actively managed funds (like MFs)

- ✓ Differenty from *index funds,* ETFs are:
  - interaction
    - investors in ETFs mostly do not trade the fund *directly*
    - “Creation and Redemption mechanism” (next)

Structure of ETF



Structure of ETF

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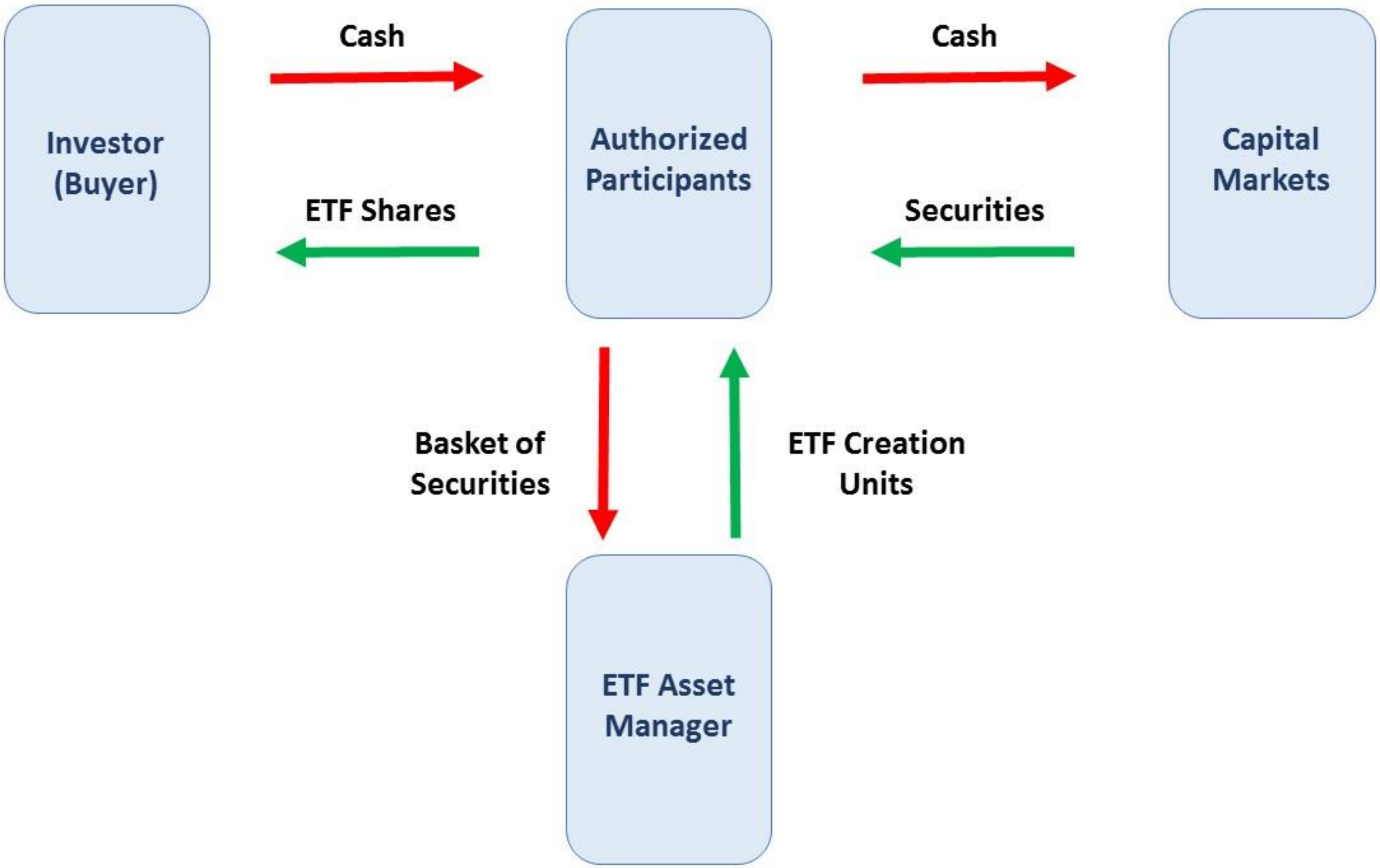


Structure of ETF

- ✓ the *ETF manager* enters into a legal contract with one/several "Authorized Participants" (APs)
  - APs have *exclusive right* to interact with ETF manager
  - typically, large financial institutions with expertise
- ✓ the ETF manager can *issue* or *redeem* shares, called creation units, with APs (in large blocks) in exchange for
  - a *basket of securities* ("in-kind" transactions)

    and/or
  - cash
- ✓ APs in turn interact with the **markets**
  - buy/sell with investors
    - ETF shares or securities making the index
  - often hold (at their own risk) some *inventory* of the ETF shares, creating further market liquidity for the ETF

Structure of ETF

Primary market of ETF shares

- ✓ Players: only ETF manager and AP’s
- ✓ The mechanism by which the shares of the ETF are adjusted in response to supply and demand is known as ***creation/redemption mechanism***
- ✓ **Creation** = **increase** in the *number of shares* outstanding
- ✓ **Redemption** = **decrease** in *the number of shares* outstanding

ETF price

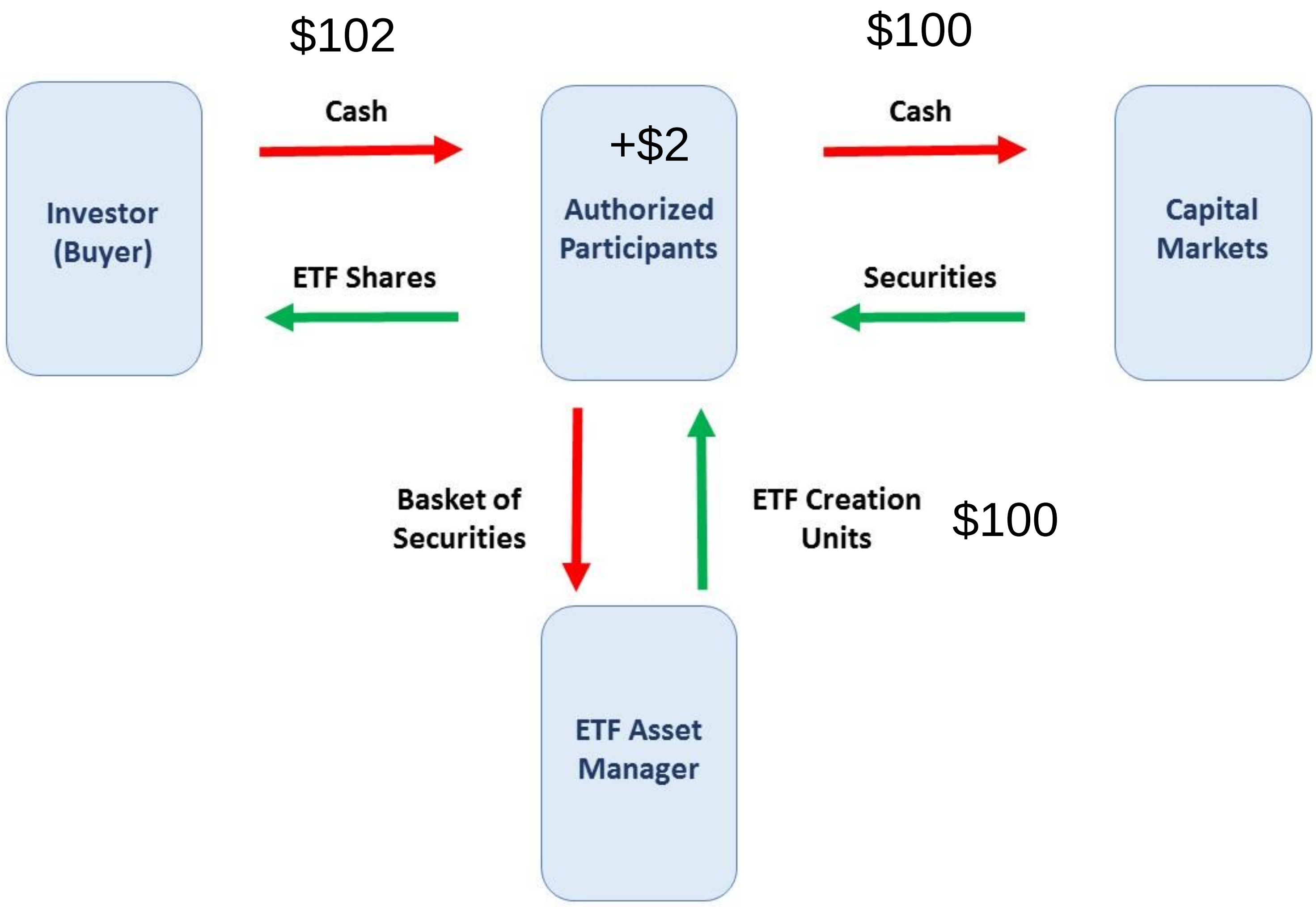
- ✓ ETF managers have to publish a NAV (like MFs)

- ✓ Arbitrage of APs
  - deviations of price from the announced NAV are **arbitraged away** by APs thanks to the creation/ redemption mechanism. How?

  - Ex. 1: ETF is traded *at premium* wrt AP's estimate of the value ($NAV_t > NAV^e$; 102 Vs 100). The AP:
    - buys basket of securities making index from investors at NAV^e
    - exchange it with ETF manager to get ETF shares
    - sell ETF shares to the market, to lock in infraday profits at $NAV_t \Longrightarrow NAV \downarrow$

  - Ex. 2: ETF is traded at *discount* ($NAV_t < NAV^e$). The AP:
    - buys ETF shares from the investors at $NAV_t \Longrightarrow NAV \uparrow$
    - exchange them to ETF manager to get basket of securities
    - sell the securities to market at NAV^e

ETF traded *at premium*



ETF price

- ✓ The creation/redemption mechanism thus works through *arbitrage* to help keep the **price of an ETF <u>close</u> to the intrinsic value** of its holdings

- ✓ Market data vendors calculate and publish the NAV based on past prices and provide an **Intraday *Indicative* Value** <u>*every 15 seconds*</u>!

- ✓ The *intraday tradability* of ETFs allows to have the price of the fund determined by the market through the interaction of buyers and sellers

- ✓ In contrast, for MFs liquidity is offered <u>only at the close and only at NAV</u>

Hedge Funds

Hedge Funds are a special type of Mutual Funds

- do not hedge, but they do **speculation**

- they often pursue risky strategies that yield *high returns* in both falling and rising markets

- Only wealthy investors allowed (by law):
  - high minimum investment (min. wealth: $1M)
  - high fees: typically, 1% of assets plus 20% of profits (or even 50%!)
  - annual income >$200,000 and net worth (excluding house) >$1M

- Lightly regulated wrt MFs:
  - idea is that the **rich can take care of themselves** (they do not need the government to tell them not to gamble)
  - can do short-selling ➡ highly levered
  - derivatives trading
  - program trading

Hedge Funds

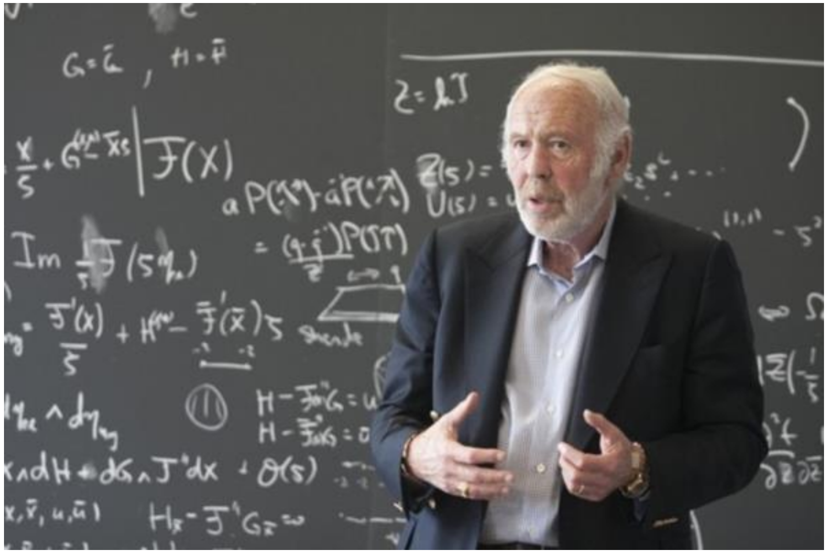
- ✓ Hedge funds employ *sophisticated* trading strategies
  - they try to take advantage of unusual spreads between security prices

- ✓ Ex: it can be that a 29.5-year US Treasury bond seems cheap relative to 30-year Treasuries. The managers figure that the value of the two bonds *would converge* over time: similar maturity. Strategy:
  - sell short ($2bn) of the 30-year and go long ($2bn) on the 29.5-year
  - 6 months later, reverse these transactions:
    sell 29.5-year and buy 30-year
  - if prices do converge realize a profit: that from $P_{29.5\ y}$ ↑ should cover loss from $P_{30\ y}$ ↓ and repay borrowing costs
  - if they do not converge «enough»: big loss

- ✓ Everything amplified by using lots of **leverage**: borrow a lot so ROE is very high

Renaissance Technologies

Sophisticated hedge funds run by mathematicians



James Simons, Renaissance Technologies

- Renaissance flagship's fund Medallion has had a 60% average return 1994-2016
- Worst year was 21%. Made 98% return in 2008.
- James Simons himself made $2.5bn of income (!) in 2008
- He is considered "the most successful *hedge fund* manager of all time"

Mini-Case: The LTCM Debacle

- ✓ Long Term Capital Management (*LTCM*) was a *hedge fund* run by John Meriwether, and its board included Nobel Laureates (Myron Scholes and Robert C. Merton)
- ✓ It recorded after-fee returns in excess of 30% for the first several years
- ✓ It then took *bets* that went the wrong way (*prices did not converge* after 1997 Asian Crisis & Russia default)
- ✓ Leverage: in 1998 $4.72bn net worth, $129bn assets ... Regulator had to bail it out. Its failure could have been devastating for the U.S. economy
- ✓ You can read more about it here if interested

Conflicts of Interests (optional)

- ✓ investment advisers have incentive to increase their income (through fees) to the detriment of shareholders' return
- ✓ monitoring by directors (in turn appointed by investors) is not sufficient

2 Relevant cases of substantial conflicts of interest

- ✓ *Late trading:* practice of allowing trades at the 4:00pm price of that day after 4:00pm to trade (not tomorrows price)
  - if relevant news became public after 4:00pm, a trader could make a **certain gain**
  - late trades are illegal but there are some exceptions
- ✓ *Market timing:* MFs calculate NAV using latest released prices; but due to time zone differences, these may be already many hours old (and fail to contain relevant information)
  - Ex: US fund that holds Japanese stocks. Japan stock market closes 9h before US market. NAV of US fund uses closing prices of Japanese market, which do not contain new information that as released after Japanese market closed